

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented high-throughput, low-latency data replication algorithms for in-memory distributed services, ensuring sub-millisecond latency at scale.
- Developed robust authentication and authorization protocols for distributed systems, managing access control for sensitive data stores.
- Contributed net new features and performance optimizations to open-source Valkey, collaborating with the OSS community to influence the project's direction.
- Built and scaled distributed services on **AWS**, focusing on horizontal and vertical scaling strategies for in-memory data stores like **Redis OSS** and **Memcached**.
- Wrote production runbooks and internal documentation that the customer onboarding team now uses to bring new clients live in ~10 days instead of 4+ weeks.
- Containerised the feature-pipeline services with **Docker** and deployed them to a **Kubernetes** cluster (EKS), wiring up Helm chart configuration and basic resource limits for predictable rollout.
- Participated in weekly operational gap investigations, proactively root-causing and resolving issues to improve system reliability and reduce future paging events.

SKILLS

Cloud Platforms – AWS, EC2, S3, Lambda, CloudWatch

Databases – Redis OSS, Memcached, Valkey, PostgreSQL, MySQL, Oracle

Programming Languages – Java, Python, SQL, C++, Rust

Core Concepts – Distributed Computing, Data Storage, System Performance, Memory Management, Parallel Computing, NoSQL

Developer Tools – Docker, Kubernetes, Jenkins, Git, Jira, Confluence

Data Processing – ETL, Data Pipelines, Data Warehousing

PROJECTS

Distributed In-Memory Data Store Scaling

- Engineered sharding algorithms for a distributed in-memory data store, improving data distribution efficiency by **30%** across a 50-node cluster.
- Implemented a novel data replication strategy using **Java Spring Boot** and asynchronous messaging, reducing data consistency lag to under **50ms**.
- Developed automated deployment scripts using **Jenkins** and **Docker** to provision and configure new nodes, cutting deployment time by **40%**.

Real-time Data Ingestion and Processing Pipeline

- Built a **Python**-based ETL pipeline to ingest and process real-time transaction data from multiple sources, handling approximately **3,500 events/sec** during peak hours.
- Designed and implemented REST APIs using **FastAPI** for downstream services to access processed data, ensuring p95 latency remained below **180ms**.
- Integrated the pipeline with **PostgreSQL** for persistent storage of processed records, supporting ad-hoc querying for operational monitoring and reporting.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)